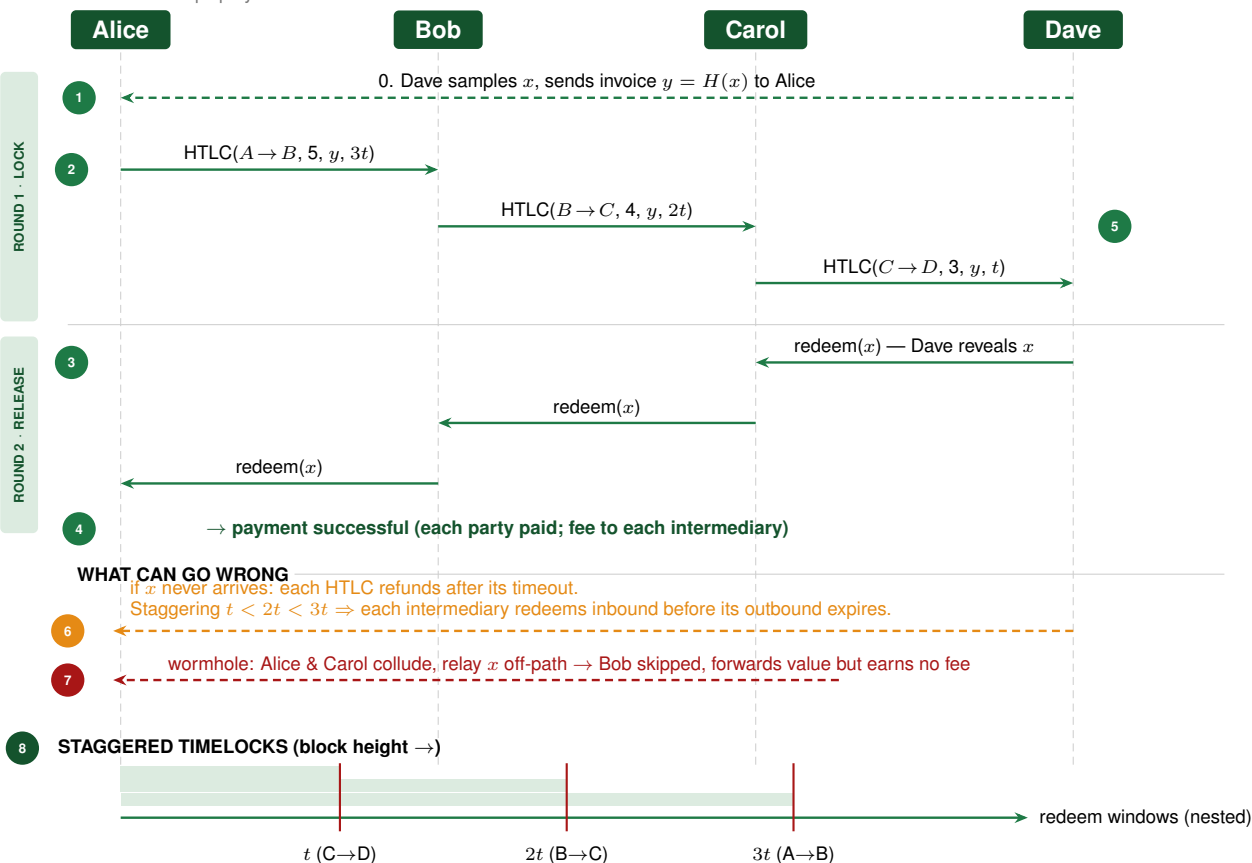


Multi-hop payment: Alice → Bob → Carol → Dave

annotated with multihop.spthy lemmas — see rail →



LEMMAS · multihop.spthy

- Invoice & preimage authentication**
Invoice_Authenticates_Settlement · Preimage_Secret_Until_Released
Forged_Invoice_Requires_Key_Compromise · Invoice_Released_Once ✓
- Lock causality (forward needs previous)**
Forward1_Requires_Offer · Forward2_Requires_Forward1
HTLC_On_Opened_Channel ✓
- Release causality (redeem needs preimage)**
Fulfill_Requires_Forward2 · Claim_Requires_Release
Settle_Requires_Receiver_Release ✓
- Atomicity + non-vacuity**
Payment_Atomicity_Under_Liveness ✓
Settle_Excludes_Sender_Refund ✓ (no refund after settle)
Multihop_Payment_Possible ✓ (witness)
- Value & fee conservation**
EndToEnd_Value_Conservation · Fee_Conservation_Hop1/2
Fee_Strictly_Positive_Hop1/2 · Receiver_Paid_Invoice_Amount ✓
Fees_Charged_On_Path_Possible ✓ (witness: both hops earn)
- Refund / grieving**
Refund_Requires_Timeout · Loss_Requires_Inaction ✓
Honest_Intermediary_Refunds ✓ (reachable refund/DoS)
- Wormhole fee theft**
Wormhole_Fee_Theft_Reachable ✓
Wormhole_Steals_Exactly_The_Fees ✓ (theft = path fees)
- Intermediary safety (under liveness T2b)**
Intermediary_Never_Loses_Under_Liveness ✓
Timeout_Race_Blocked ✓ (race blocked with T2b)

✓ = machine-checked in multihop.spthy (43/43).
green = all-traces safety · orange = benign/reachable
red = attack shown reachable (by design, no PTLC).

Channel-layer lemmas (open/update/close/punish)
live on the Alice ↔ Bob figure.